Multimedia Signal processing

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September, 3 2012

1. **Question (8 pts)**:
   Consider a continuous source X, whose p.d.f. is given by

$$p_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0 \tag{1}$$

   Set $\lambda = 1$ and consider the following sequence of symbols

$$\{0.1, 0.0, 0.5, 2, 0.2\} \tag{2}$$

   (a) Design a quantizer based on compander-expander. Use 2 bits to represent each symbol. Show the equivalent non-uniform quantizer in the original signal domain.

   (b) Quantize the sequence of symbols using the quantizer above. Compute the reconstructed sequence at the decoder and the MSE distortion.

   (c) Quantize the sequence of symbols using a uniform scalar quantizer in the range $[0, 2]$. Compute the reconstructed sequence at the decoder and the MSE distortion.

2. **Question (5 pts)**
   Discuss the properties of speech signals, both in the time and in the frequency domain.

3. **Question (5 pts)**
   (a) Illustrate the block diagram of a transform coder.

   (b) Discuss the design of the optimal transform and quantizer based on the statistics of the input source.

4. **Questions (5 pts - each answer can be either TRUE or FALSE)**
   (a) Let $R(D)$ denote the rate-distortion function of a discrete signal without memory.
   - [T] [F] $R(0) = \sigma_x^2$
   - [T] [F] $R(D_1) > R(D_2)$, if $D_1 > D_2$
   - [T] [F] $R_{TC}(D) < R_{PCM}(D)$, where $R_{TC}(D)$ and $R_{PCM}(D)$ are, respectively, the operational rate-distortion curves of transform coding and scalar quantization.
   - [T] [F] If $x \in \mathcal{U}(0,1)$, $D(R) = \frac{1}{12}2^{-2R}$

   (b) Consider an image coding system
   - [T] [F] Entropy coding of transform coefficients is invertible.
   - [T] [F] The coding efficiency decreases when the block size is increased.
   - [T] [F] The coding efficiency of zonal quantization is higher than that of threshold quantization

   (c) Consider a video coding system.
   - [T] [F] The encoder computes a copy of the video frames identical to that available at the decoder.
   - [T] [F] The average PSNR of prediction residuals of full search motion estimation is smaller than (or equal to) that of fast search.
   - [T] [F] Motion vectors are lossy coded.

5. **Question (Matlab - 10pts)**

   A wireless sensor network is designed to perform speech acquisition. Each sensor records speech samples from the environment, encodes and transmits them to a central controller, where speech is analyzed. Assuming that:

   - To maximize recognition accuracy, the distortion introduced by lossy coding (SNR) should be greater than 15 dB.
   - Each sensor consumes 20 nJ to transmit 1 bit of data.

   Compute the energy dissipated to transmit the first 2 seconds of the file *speech.wav* under the accuracy constraint, when a sensor uses:

   (a) PCM coding.

   (b) DPCM coding.

   (c) PCM coding + Entropy coding (in this case assume to use exactly $H(X)$ bits per sample, where X denotes the source of quantized speech samples (assume a discrete memoryless source).